

Box-counting dimension of the graph of fractal interpolation f :

$N \geq 2$, $\{(x_n, F_n) : n = 0, 1, 2, \dots, N\}$ data set

$$w_n \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} a_n & 0 \\ c_n & d_n \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} e_n \\ f_n \end{pmatrix}, \quad n = 1, 2, \dots, N$$

$$0 \leq d_n < 1$$

IFS : $\{\mathbb{R}^2; w_n, n = 1, 2, \dots, N\}$

G = unique attractor of this IFS.

G is the graph of fractal interpolation

f if $\sum_{n=1}^N d_n > 1$ and the interpolation points do not all lie on a single straight line,

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then the box-counting dim. of G is the unique real no. D satisfying

$$\sum_{n=1}^N d_n a_n^{D-1} = 1.$$

Special case : $x_i = x_0 + i \left(\frac{x_N - x_0}{N} \right), i = 0, 1, 2, \dots, N$

Then $a_n = \frac{1}{N}$ for $n = 1, 2, \dots, N$

$\therefore D$ is given by

$$\sum_{n=1}^N d_n \left(\frac{1}{N} \right)^{D-1} = 1$$

$$\text{i.e. } N^{D-1} = \sum_{n=1}^N d_n$$

$$\Rightarrow D = 1 + \frac{\ln \left(\sum_{n=1}^N d_n \right)}{\ln N}$$

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Hence, by choosing d_n 's appropriately, we can make the dimension D to be any real number in $(1, 2)$.

Dimensions of graphs of functions

Prop: Let $f: [0, 1] \rightarrow \mathbb{R}$ be continuous. Let $0 < \delta < 1$, and let $m = \lceil \frac{1}{\delta} \rceil$. Then, if N_δ is the number of squares of the δ -mesh that intersect $\text{graph}(f)$, then

$$\frac{1}{\delta} \sum_{i=0}^{m-1} R_f[i\delta, (i+1)\delta] \leq N_\delta \leq 2m + \frac{1}{\delta} \sum_{i=0}^{m-1} R_f[i\delta, (i+1)\delta],$$

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where $R_f[t_1, t_2] = \sup_{x, y \in [t_1, t_2]} |f(x) - f(y)|$

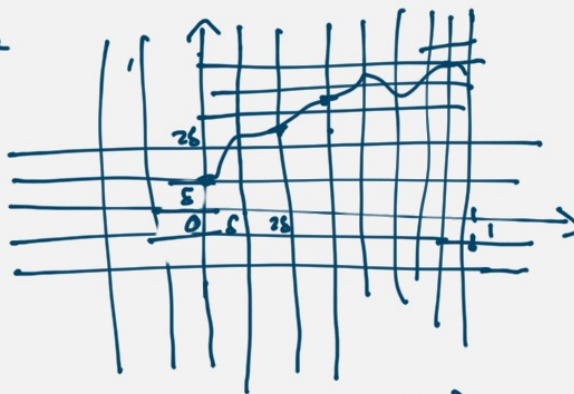
Proof:

The no. of squares in the δ -mesh in the column above $[i\delta, (i+1)\delta]$ that intersect $\text{graph}(f)$ is at most

$$2 + \frac{R_f[i\delta, (i+1)\delta]}{\delta}$$

(This does not need continuity of f)

$$\begin{aligned} \therefore N_\delta &\leq \sum_{i=0}^{m-1} \left(2 + \frac{R_f[i\delta, (i+1)\delta]}{\delta} \right) \\ &= 2m + \frac{1}{\delta} \sum_{i=0}^{m-1} R_f[i\delta, (i+1)\delta] \end{aligned}$$



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Also, since f is continuous, using intermediate value theorem,

$$N_\delta \geq \sum_{i=0}^{m-1} \frac{R_f[i\delta, (i+1)\delta]}{\delta}$$

This proves the proposition.

Theorem: Let $f: [0,1] \rightarrow \mathbb{R}$ be continuous.

(a) Suppose $|f(t) - f(u)| \leq c|t-u|^{2-s} \forall t, u$,

where $c > 0$ and $1 \leq s \leq 2$.

Then $H^s(\text{graph } f) < \infty$ and
 $\dim_H(\text{graph } f) \leq \underline{\dim}_B(\text{graph } f) \leq \overline{\dim}_B(\text{graph } f) \leq s$.

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(b) Suppose $\exists c > 0, \delta_0 > 0$ and $1 \leq s < 2$ such that for each $t \in [0,1]$ and $0 < \delta \leq \delta_0$,
 $\exists u \in [0,1]$ with $|t-u| \leq \delta$
 and $|f(t) - f(u)| \geq c\delta^{2-s}$.

Then $\underline{\dim}_B(\text{graph } f) \geq s$.

Proof: (a) Since $|f(t) - f(u)| \leq c|t-u|^{2-s}$,

$$R_f[t_1, t_2] \leq c|t_1 - t_2|^{2-s}$$

$$\Rightarrow R_f[i\delta, (i+1)\delta] \leq c\delta^{2-s}$$

$$\therefore \text{By the prev. prop., } N_\delta \leq 2m + \frac{1}{\delta} m c \delta^{2-s}$$

$$\leq 2\left(1 + \frac{1}{\delta}\right) + \left(1 + \frac{1}{\delta}\right) c \delta^{1-s}$$

$$\leq c_1 \delta^s \text{ for some } c_1 \text{ indep. of } \delta$$

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$$\Rightarrow \dim_B(\text{graph } f) \leq s$$

and also $H^s(\text{graph } f) < \infty$.

$$(b) \quad R_f[i\delta, (i+1)\delta] \geq c \delta^{2-s}$$

$$\therefore N_\delta \geq \frac{1}{\delta} m c \delta^{2-s} \geq \frac{1}{\delta} \cdot \frac{1}{\delta} c \cdot \delta^{2-s} = c \delta^{-s}$$

$$\Rightarrow \underline{\dim}_B(\text{graph } f) = \lim_{\delta \rightarrow 0^+} \frac{\ln N_\delta}{-\ln \delta} \geq s.$$

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Example (Weierstrass function)

Let $\lambda > 1$ and $1 < s < 2$.

Define $f: [0, 1] \rightarrow \mathbb{R}$ by

$$f(t) = \sum_{k=1}^{\infty} \frac{(s-2)^k}{\lambda^k} \sin(\lambda^k t).$$

Then, provided λ is large enough,

$$\dim_B(\text{graph } f) = s.$$

Proof: Given $0 < h < \lambda^{-1}$, let N be the integer
st. $\lambda^{-(N+1)} \leq h < \lambda^{-N}$

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Then,

$$\begin{aligned}
 |f(t+h) - f(t)| &= \left| \sum_{k=1}^{\infty} \frac{(\lambda^{-2})^k}{\lambda} \left(\sin\left(\frac{k}{\lambda}(t+h)\right) - \sin\left(\frac{k}{\lambda}t\right) \right) \right| \\
 &\leq \sum_{k=1}^N \frac{(\lambda^{-2})^k}{\lambda} \left| \sin\left(\frac{k}{\lambda}(t+h)\right) - \sin\left(\frac{k}{\lambda}t\right) \right| \\
 &\quad + \sum_{k=N+1}^{\infty} \frac{(\lambda^{-2})^k}{\lambda} \left| \sin\left(\frac{k}{\lambda}(t+h)\right) - \sin\left(\frac{k}{\lambda}t\right) \right| \\
 &\leq \sum_{k=1}^N \frac{(\lambda^{-2})^k}{\lambda} \lambda^k h + 2 \sum_{k=N+1}^{\infty} \frac{(\lambda^{-2})^k}{\lambda} \\
 &= h \sum_{k=1}^N (\lambda^{-1})^k + 2 \sum_{k=N+1}^{\infty} (\lambda^{-2})^k \\
 &\leq h \frac{\lambda^{(s-1)N}}{1 - \lambda^{1-s}} + \frac{2(\lambda^{-2})^{N+1}}{1 - \lambda^{-2}}
 \end{aligned}$$

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Using $\lambda^{-(N+1)} \leq h < \lambda^{-N}$, we get

$$|f(t+h) - f(t)| \leq c h^{2-s} \text{ for some constant } c \text{ indep. of } h.$$

$\therefore$ By (a) of the prev. thm,

$$\dim_B(\text{graph } f) \leq s.$$

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